

Math 563: Mathematical Genetics

Instructor: Benjamin Ridenhour

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Office Hours: MWF 2:30–3:30pm (or by appt.)

Office: Brink Hall 421

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Class Hours: MW 5:10–6:25pm

Class Room: TLC 248

Course Description

Math 563 is a graduate course meant to expose students to important mathematical models used in the field of genetics. Because this is a mathematics course, the focus will primarily be on modeling (such as forward-time models) but basic concepts in genetics and evolution will also be introduced. The role of mathematical models in genetics has a long and storied history involving many famous scientists such as Fisher, Wright, Price, Nei, Hamilton, Lande, and many others. These models span statistics, differential equations (stochastic and deterministic), probability, algebra, and other mathematical fields, thus students must be prepared to encounter varied forms of mathematics. Many of the models we will survey in the course form the basis of our modern understanding of the evolutionary process.

The structure of the class is as follows. We plan to achieve the goals of the course through a series of instructor led “workshops” and lectures on topics in mathematical genetics; student led final presentations will occur during the last few weeks. One or two readings will usually be assigned per topic: one for background and a second for discussion.

For workshops, an instructor will briefly motivate the topic, outline the analyses, and summarize the main results. The class will then work through the reading(s) as a group, stopping to clarify steps, methods, or concepts that are ambiguous. Ideally, students who understand a step/method/concept will do the clarifying. Of course, the instructor will help regulate the flow of discussion and contribute lucid explanations as necessary! The instructor will also suggest homework exercises that all students are expected to attempt. For this workshop format to succeed, it is critical that every student works to understand the readings prior to class. If, in the course of reading the material, you don’t understand something (e.g., a term, concept, manipulation, or technique) make a note of it and go through the remaining material as best you can. Be sure to bring your notes to class.

Student-led final presentations will cover topics chosen by students (topics—which may involve original research—must be pre-approved by the instructor). Presenters should prepare handouts and readings that their fellow students will read prior to the discussion.

AI Usage

For better or worse, you now have the world's most powerful "tutor" at your fingertips via various AI platforms. If used properly, such platforms can potentially help you understand how to solve the complex problems that arise in mathematical genetics. That being said, I would strongly warn against reading solutions provided by AI and then thinking you yourself understand how to solve such problems. Without actively challenging yourself, you will not learn the material. Any usage of AI in performing homework assignments or in creation of your final presentation should *extremely limited and cited* if you do use some platform. In general, I ask that you refrain from using AI for Math 563. As I mentioned before, such tools can provide a valuable aid to your understanding of the materials in this course, but it is also easy to convince yourself you know the material when you have not actually properly solved the problems on your own.

Required Materials

- *Population Genetics: A Concise Guide*. 2nd edn. by J. Gillespie. The John Hopkins University Press: Baltimore.

Prerequisites/Corequisites

Prerequisites: Multivariate calculus (Math 275), Genetics (Biol 310), Statistical analysis (Stat 431), Probability (Math 451), or instructor permission.

Topics

We will spend roughly 2 weeks (4 class periods) on each of the following topics:

1. Hardy Weinberg Equilibrium & Genetic Drift
2. Natural Selection
3. Two-locus Selection & Non-random Mating
4. Quantitative Genetics
5. Coalescent Theory: Mathematical Underpinnings
6. Coalescent Theory: Statistical Inference
7. Effects of Population Structure, Effective Population Size, & MCMC
8. Final presentations by students

Course Objectives/Learning Outcomes

Successful students will:

1. Gain exposure to, and experience with, major topics and methods in mathematical genetics.
2. Survey classical and contemporary literature in mathematical genetics.

Final Exam Date

No final exam will be given.

Grading Policy

A typical grading scale will be used (i.e., $\geq 93\% = A$, $(93,90\%] = A-$, $(90,87\%] = B+$, ...). I reserve the right to curve the scale dependent on overall class scores at the end of the semester. Any curve will only ever make it easier to obtain a certain letter grade. The grade will count the assessments using the following proportions:

- 35% of your grade will be determined by homework.
- 35% of your grade will be determined by your final presentation.
- 20% of your grade will be determined by the class participation.
- 10% of your grade will be determined by peer reviews of your final presentation.